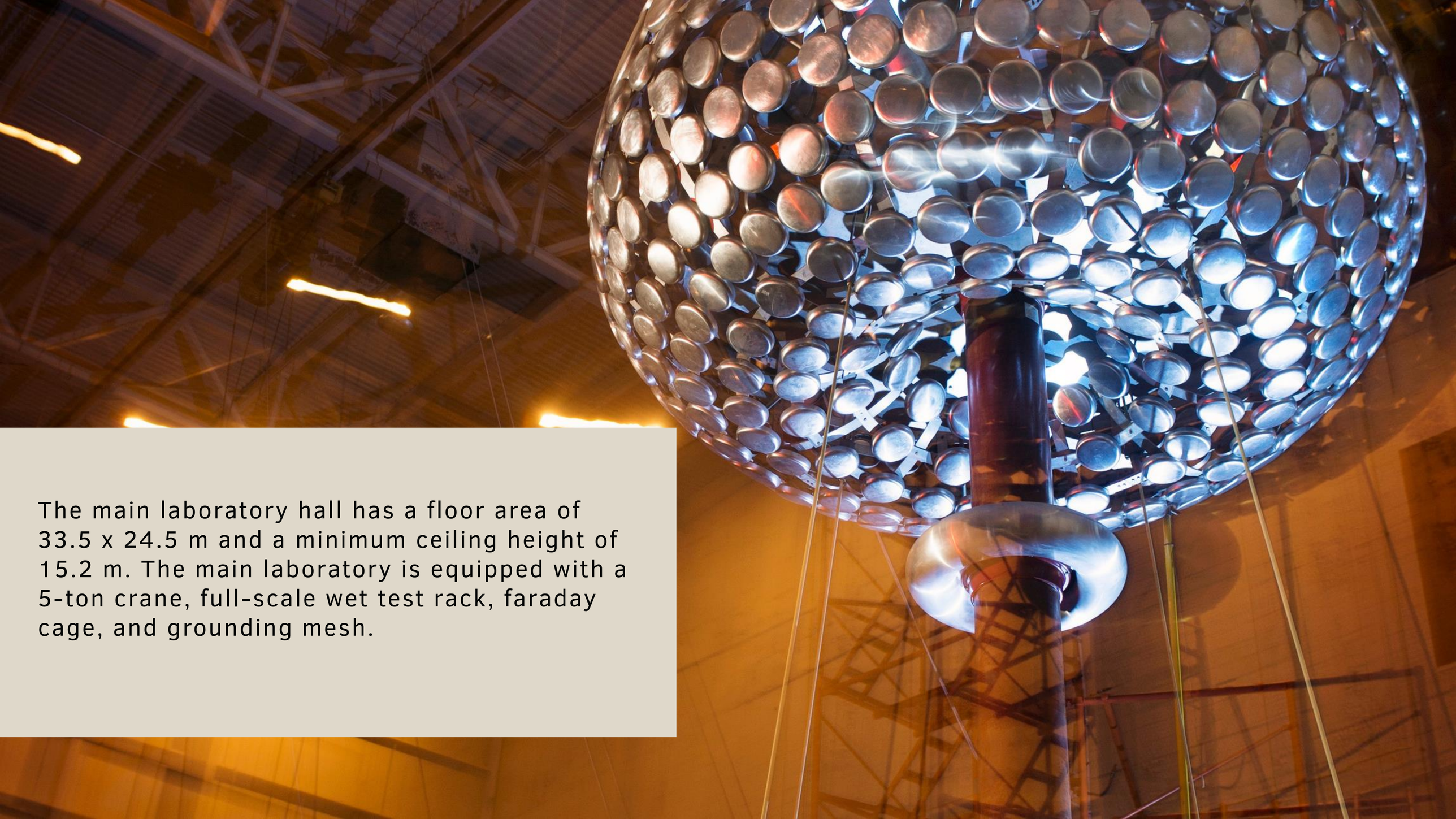
HIGH VOLTAGE LABORATORY AND TEST

High Voltage laboratory



Description and history of the laboratory

The Paul B. Jacob High Voltage Laboratory serves as an independent, non-industrial university center for Mississippi State University. The laboratory was built in 1977 and is the largest university-owned high voltage laboratory in North America. The laboratory focuses on high voltage research, evaluation, and education. The laboratory has been designed to meet the evaluation needs of the industry and enables academic research in the field of high voltage engineering.



The main laboratory hall has a floor area of 33.5 x 24.5 m and a minimum ceiling height of 15.2 m. The main laboratory is equipped with a 5-ton crane, full-scale wet test rack, faraday cage, and grounding mesh.



Main testing equipment and capabilities

- 3000 kV (56 kJ) impulse generator
- 1000 kV (1 MVA) 60 Hz transformer
- 400 kV DC supply
- 250 kV 60 Hz transformer
- 400 kV portable impulse generator
- 400/200/100 kV multi test set (impulse/DC/AC)
- Programmable surge generator (EMC)
- Multiple dielectric test sets
- Climate chamber (-73°C ... +175 °C, 27 cubic ft workspace)
- Porosity test setup (max. 5000 psi)
- Custom accelerated ageing chamber
- 3 tracking and erosion systems (RWDT)
- Standard wet test facility

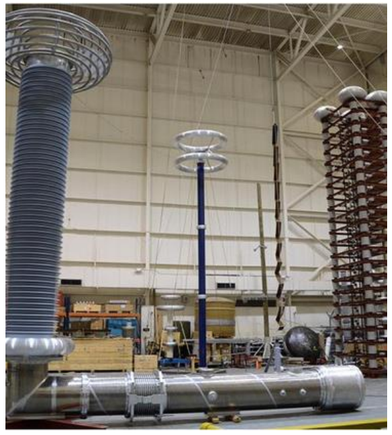
Which tests are being conducted in the lab?

Recent Testing Projects

- Design, qualification, periodic electrical test evaluation of insulating aerial devices up to 500 kV
- Electrical evaluation of high voltage polymer and porcelain insulators under dry and wet condition
- Tracking and erosion tests of polymer insulators and cutouts
- Measurement of radio influence voltage (RIV) and corona on line and station insulators, switchgear, and hardware up to 500 kV
- Electrical performance evaluation of transmission and distribution arresters
- Lightning, switching impulse evaluation of transmission transformers
- Underground cable dielectric evaluation
- Lightning, switching impulse evaluation of distribution and transmission switchgear assemblies
- Prototype, design, and quality conformance assessment of insulators
- Live line accessories - degree of long-term degradation under continuous stress
- Performance evaluation of fiberglass cross arm
- Dielectric withstand evaluation of silicone cutouts

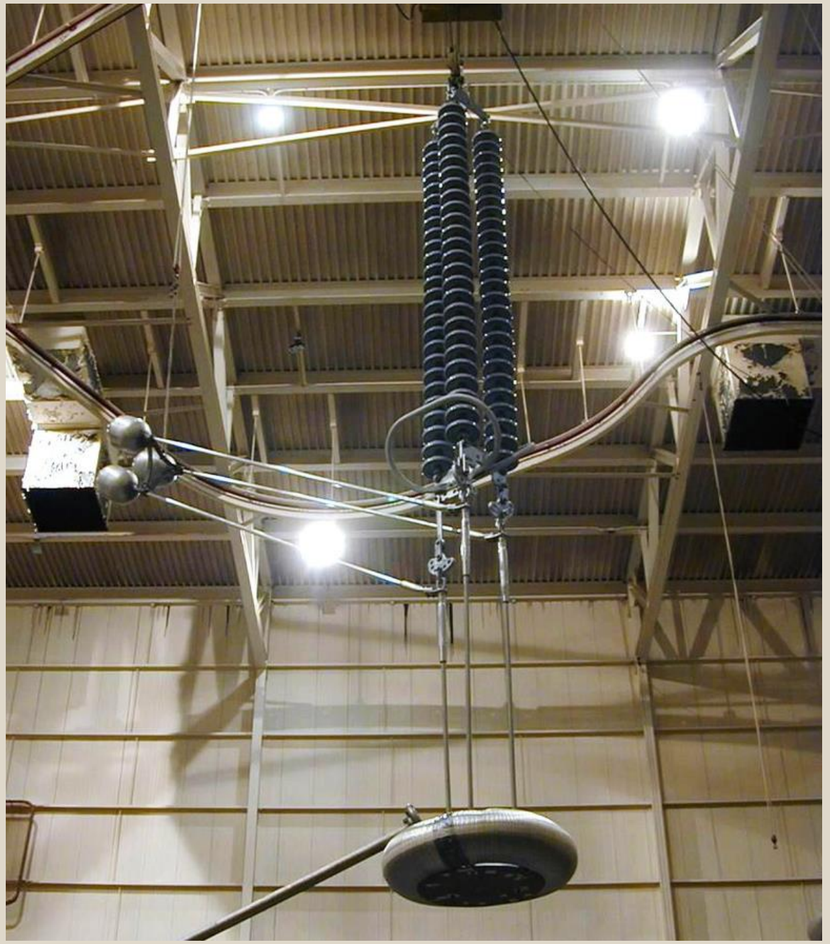






Insulator Testing Projects

- Electrical Degradation of Photovoltaic Modules Caused by Lightning Induced Voltage
- Electrical Degradation of XLPE and EPR High Voltage Cable Insulation
- Online Monitoring of MV Power Cable Insulation Condition
- AC Withstand Voltage, Lightning and Switching Critical Impulse Voltage (CFO),
- Visual Corona and Radio Influence Voltage (RIV) on Transmission Line Assemblies up to 500 kV.
- AC Withstand Voltage, Lightning and Switching Impulse Withstand Voltage under Dry and Wet conditions on 550 kV SF6/Air Polymer Bushing and 550 kV SF6 Bus Sections.
- AC Withstand Voltage, Lightning and Switching Impulse Withstand Voltage, Chopped Wave Withstand Voltage, Corona and Radio Influence Voltage (RIV) on 550 kV RLSwitcher, with SF6 Insulation.
- AC Withstand Voltage, Lightning and Switching Critical Impulse Voltage (CFO), Visual Corona and Radio Influence Voltage (RIV) on Switchgears up to 500 kV.
- Accelerated Aging of Wires under Pulse Frequency and High Temperature









Lightning Protection Projects

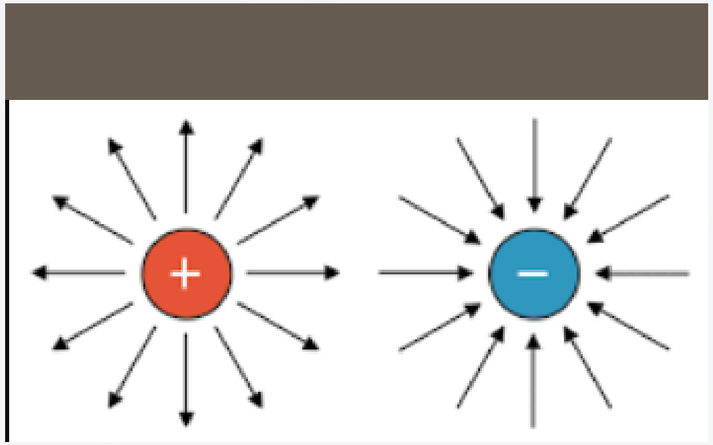
- Lightning Performance of Wind Turbine Blades
- Emission Current from Static Dissipation Elements
- Electrical Strength of Distribution Line Elements Under Steep Front, Short Duration Pulses
- Lightning Performance of Aerostat
- Lightning Protection of Naval Ships
- Laser Triggered System for Lightning Discharges
- Franklin Rod Lightning Protected Zone
- Early Streamer Emission Air Terminals
- Lightning Protection of Electrical Power Systems
- Evaluation of Aging Process in Transmission and Distribution Arresters
- Fiberglass Distribution Line Pole
- Critical Flashover Voltage of Composite Insulation
- Air Clearances in High Voltage Substation



Which standards are followed during the tests?

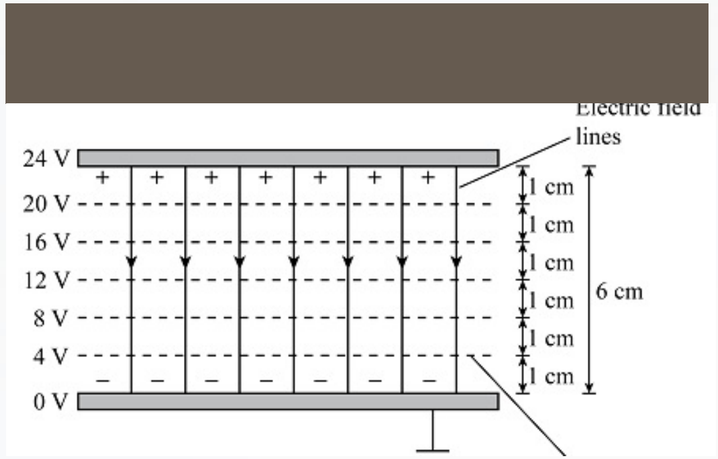
Evaluations are performed according to national and international standards as well as customer specifications.

FIELD MAPPING USING ELECTROLYTIC TANK

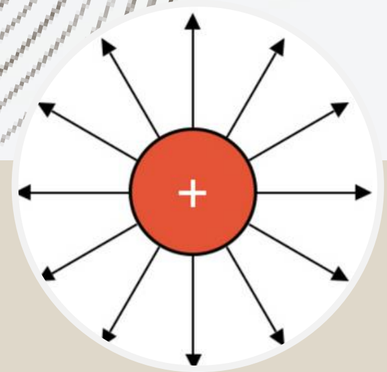


- Electric fields are fundamental to understanding the behavior of electrical systems and devices. They play a crucial role in determining the distribution of electric charges and the flow of electric currents. Field mapping is a technique used to visualize and analyze these electric fields, allowing us to study their properties.

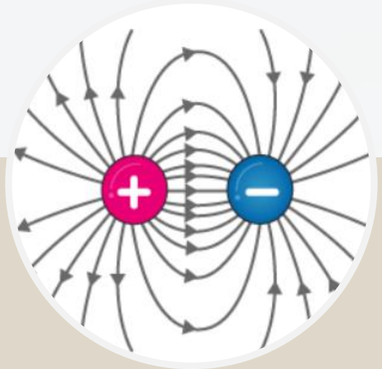
- Field mapping involves the creation of visual representation of the electric field using equipotential lines. Equipotential lines are imaginary lines that connect points in a field where the electric potential (voltage) remains constant. These lines help us understand the distribution and intensity of the electric field.



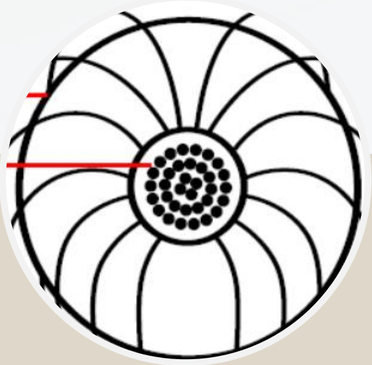
Purpose of the test, why is it needed ?



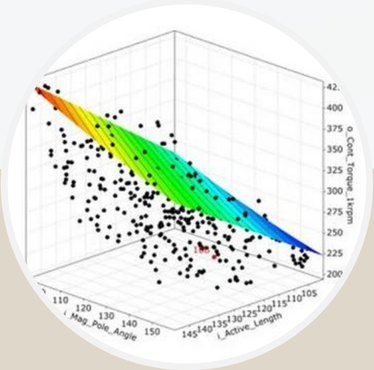
Visualizing Field Patterns



Analyzing Field Interactions



Evaluating Electric Stress



Device Optimization



Safety Assessment

EQUIPMENTS

01 1-PHASE 50HZ 230V AC SUPPLY

02 FUSE

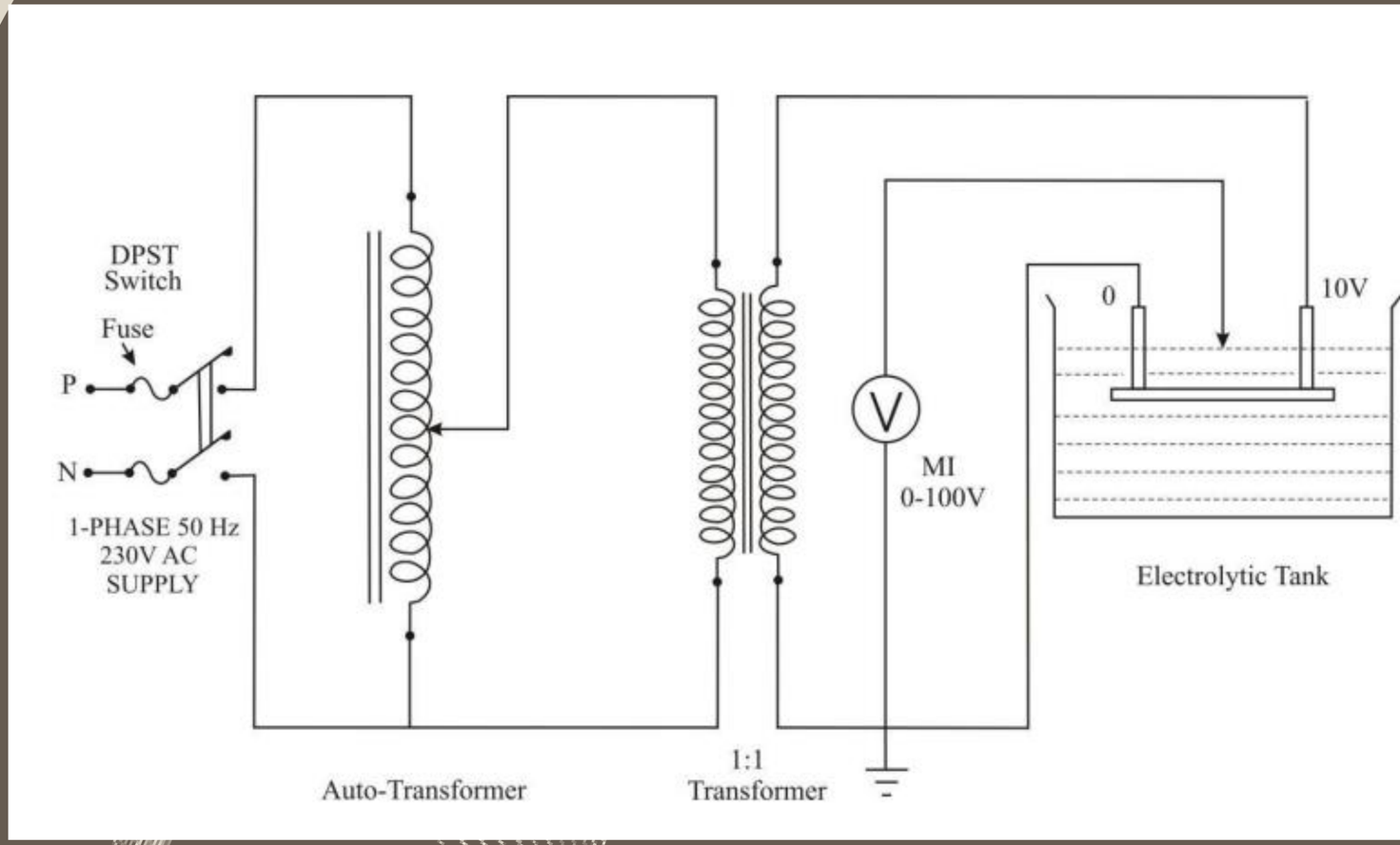
03 DPST SWITCH

04 AUTO-TRANSFORMER

05 1:1 TRANSFORMER

06 VOLTMETER

07 ELECTROLYTIC TANK



COMPONENTS



- This is the power source for the experiment. It provides a single-phase alternating current at a certain frequency and voltage. It is used to supply electrical energy to the circuit and the electrolytic tank.

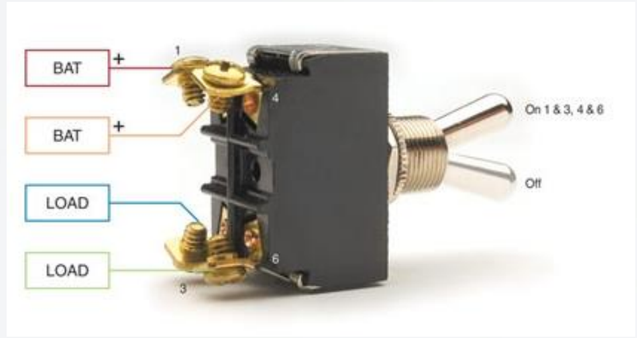
- A suitable alternative could be a generator or a variable AC power supply that can provide the required voltage and frequency.

- A fuse is a safety devide that protects the circuit from excessive current flow. It is typically placed in series with the power supply and the rest of the circuit. If the current exceeds a certain threshold, the fuse melts or blows, breaking the circuit and preventing damage to the equipment or electical components.

- In place of a fuse, a circuit breaker can be used. It serves a similar purpose by automatically openning the circuit when excessive current is detected.



COMPONENTS



- Double Pole Single Throw switch is a type of switch that can control the flow of electricity in two separate circuits simultaneously. In this experiment, the DPST switch is used to turn the power supply on or off.

- A signle pole single throw (SPST) switch can be used instead, but it will only control one circuit instead of two.

- An auto-transformer is a type of transformer with a single winding that serves both as the primary and secondary winding. It allows for variable voltage output within a specified range.
- A variac or variable transformer can be used as an alternative to adjust the voltage within the required range.



COMPONENTS

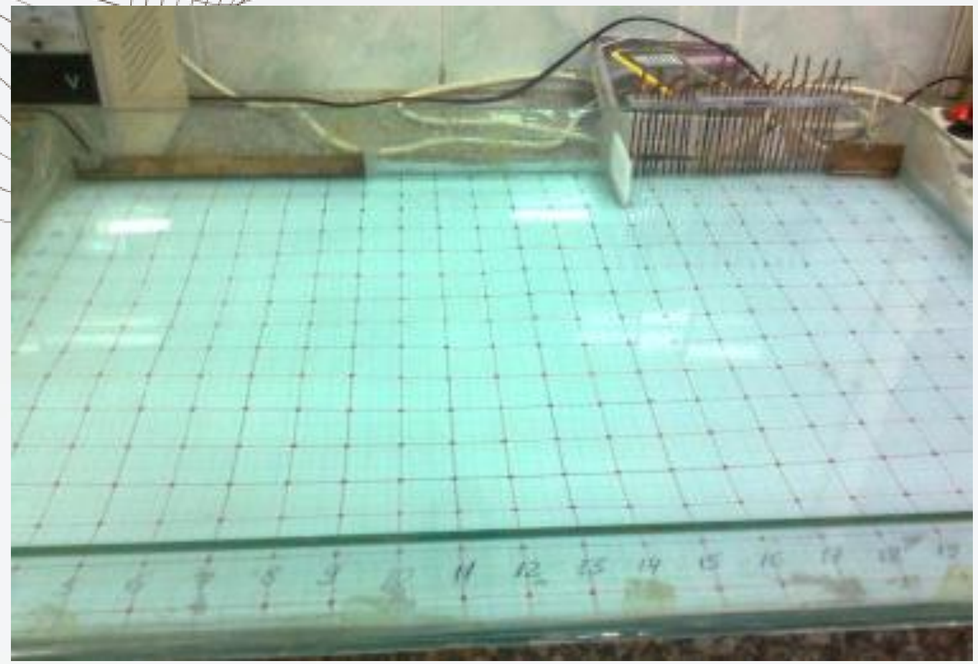


- A 1:1 transformer is a transformer with 1:1 turns ratio, meaning the primary and secondary windings have an equal number of turns.
- If the specific purpose of the 1:1 transformer is not critical for the experiment, it may be omitted or replaced with a standard transformer with appropriate voltage and current ratings.

- The voltmeter is used to measure the voltage across different points in the circuit or the electrolytic tank.
- A digital multimeter or an oscilloscope with voltage measurement capabilities can be used as an alternative. These instruments provide more accurate readings and additional features for troubleshooting and analysis.

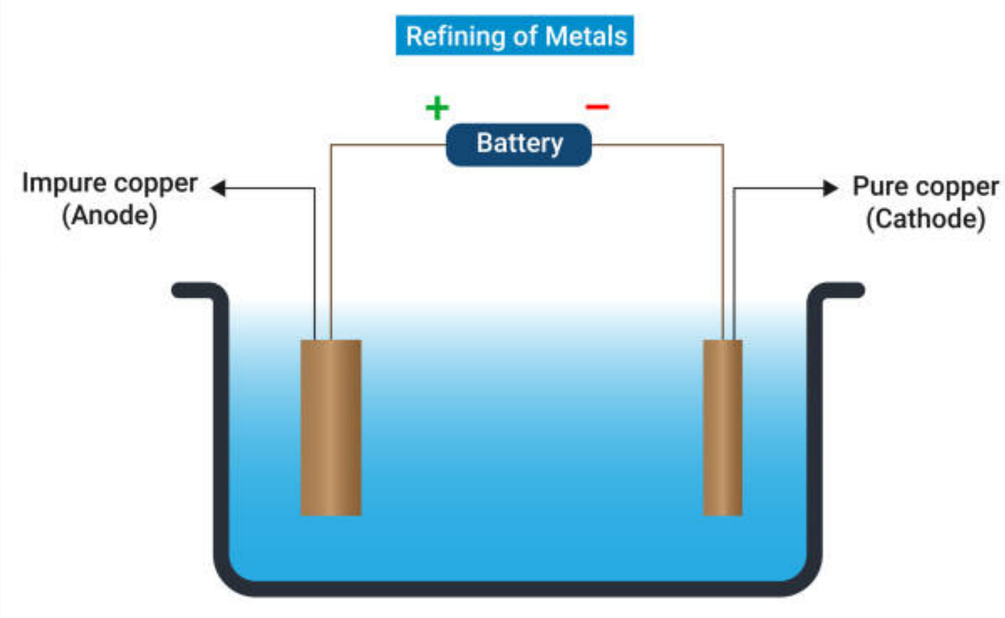


ELECTROLYTIC TANK



- The electrolytic tank is a specialized equipment used in field mapping experiments to visualize and study electric fields. It consist of a tank filled with water or an electrolyte solution that simulates the behavior of an electic field. The tank is typically made of high-quality mild steel with a protective epoxy powder coating. On top of the tank, a transparent glass plate is fixed, allowing observations and analysis of the electric field within.

- In the electrolytic tank method, the electric field is simulated by immersing electrodes or models in the tank filled with water or electrolyte solution. The models can represent various electrode configurations, such as parallel plates or co-axial cables. When a voltage is applied across the electrodes, an electric field is established in the surrounding medium, leading to the formation of equipotential lines.



Steps of the Test

Step 1 : Fill the tank with electrolyte solution

Step 2 : Place the electrodes inside the tank in the desired configuration.

Step 3 : Connect the electrodes to the power source

Step 4 : Set the power supply to the desired voltage

Step 5 : Turn on the power supply and observe the movement of the ions.

Step 6 : Map out the field lines

Step 7 : Measure and save data.

Step 8 : Repeat the experiment and compare the results with the previous experiment.

Formulas and Calculations

$$E = F\,q_o$$

E : Electric field

F : Electrical forcce

q0: Test Charge

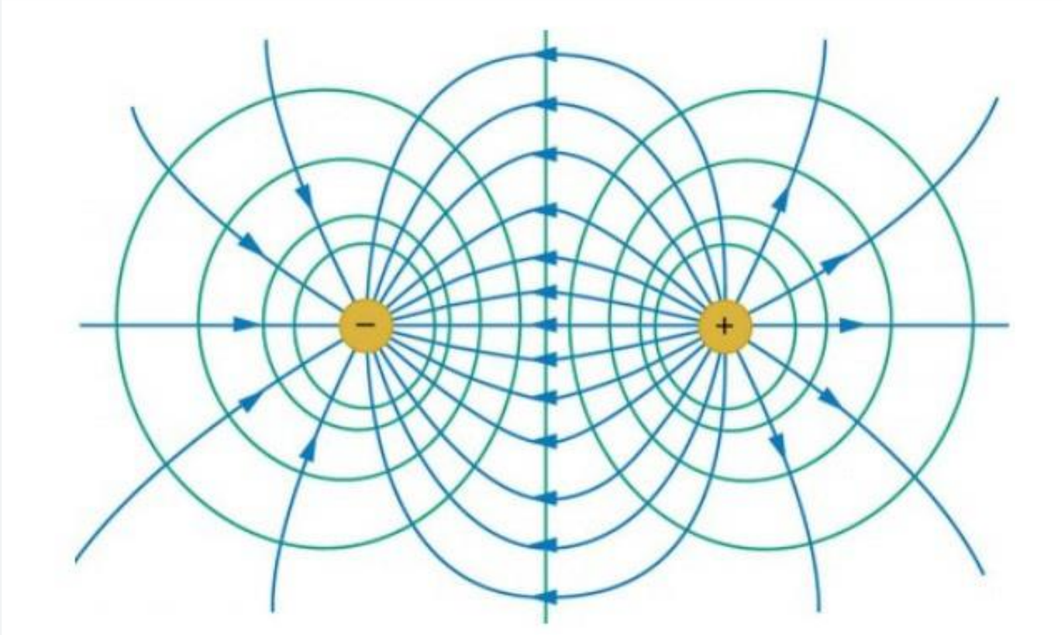
$$W = \int_A^B \boldsymbol{F} \cdot d\boldsymbol{s} = \int_A^B F\,ds\,cos\,\theta$$

W : Electrical energy

Static electric field	Stationary current field
$\vec{D} = \varepsilon\vec{E}$	$\vec{J} = \kappa\vec{E}$
$Q = \oint \vec{D}d\vec{s}$	$I = \oint \vec{D}d\vec{s}$
$C = \frac{Q}{U}$	$G = \frac{1}{R} = \frac{I}{U}$

RESULTS AND FINDINGS OF TEST

As a result of the test, traces of electric field lines are seen on a thin paper placed on the surface of the conductive liquid. The test makes it easy to analyze the properties of the electric field by visualizing electric fields. It allows to observe models and verify theoretical predictions.

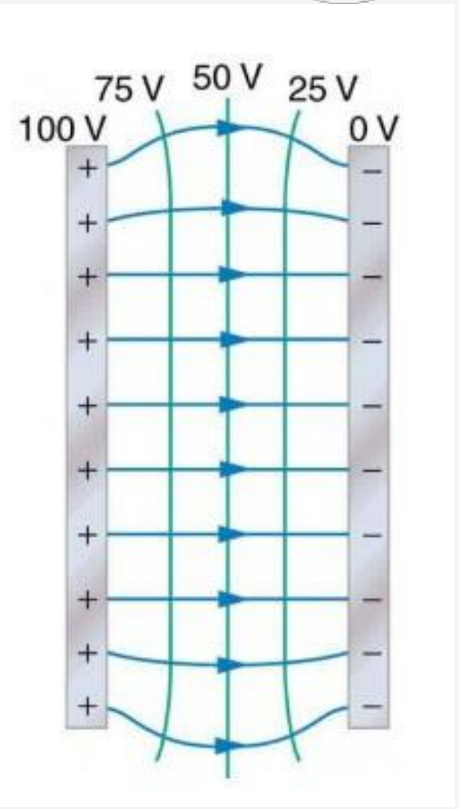


Electric field and equipotential lines with two different electrodes

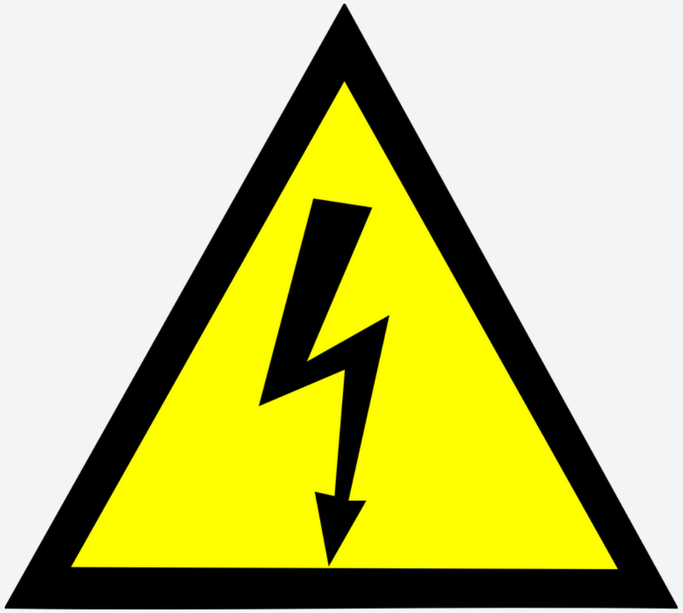
RESULTS AND FINDINGS OF TEST

Some of the test results

- Electric field lines run from the positively charged electrode to the negatively charged electrode or grounded conductors. These Lines never cross each other.
- The field lines are parallel to one another and equally spaced in homogeneous electric fields. It indicates that the area between the electrodes has a constant electric field strength.
- The intensity of the field lines represents the strength of the electric field. The closer the lines are to each other, the stronger the electric field in that place. The electric field intensity decreases as the gap between the lines decreases.
- Conductive objects or charges inside the tank can affect electric field lines. These charges can cause electric field lines to change direction and intensity.



STANDARD OF THIS TEST



Using low voltage, proper grounding and taking safety precautions by working with low risk conductive liquids are the standards of this test.

REFERENCES

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- https://www.ece.msstate.edu/high-voltage-lab/evaluationtesting-programs/
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Thank you for listening

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